AEGIS Supplementary Note: Implementation Terms and Replay Metrics

TGLTW-RMIT

Scope

This note defines implementation terms and metrics for offline replay. It contains no additional results. The main paper describes the maintained reference implementation and separately scoped historical retrieval diagnostics, not a new VQA/live-system experiment. Essential methods and limitations remain in the main paper; the project page will announce the public code release.

Implementation terms

Term	Reference implementation
Primary encoder	Tencent WeMM-Embedding-4B uses its native embedding API, L2 normalization and a default dimension of 2,560. Only checkpoint-supported MRL sizes are accepted. Index and query recipe, revision and dimension must match; archived outputs do not establish this identity.
SigLIP	Optional secondary dense ranking, disabled by default; requires matching indexed named vectors.
VLM adapter	A replaceable text/image-to-text module with a validated answer contract. No particular provider/model or model-dependent quality/latency result is disclosed in the paper.
Payload-text retrieval	Qdrant full-text filtering over available text fields. Scroll order supplies a rank; this is not a scored BM25 implementation.
RRF	Weighted, nested reciprocal-rank fusion with one-based ranks, constant 60 and all weights equal to one by default. Inner fusion combines primary dense and available payload-text rankings; outer fusion combines query/HyDE branches and optional SigLIP.
CQR / HyDE	Prompt-based contextual rewriting and hypothetical expansion. Rewriting is not a guarantee of entity preservation.
DVR / ambiguity	Distinct-video ratio over at most ten candidates, averaged equally with normalized score-margin ambiguity. The default question threshold is 0.7, not a calibrated confidence.
Clarification boost	Reply-token recall plus a 0.25 bigram/trigram bonus is capped at one. Overlap times the pre-boost maximum score is added with weight 0.5 only for question-summary IDs. Empty prior IDs leave scores unchanged.
Evidence-aware abstention	Missing media or invalid VQA output yields UNKNOWN/N/A; model errors are recorded separately. Valid JSON and a real frame do not guarantee a correct answer. Generation uses bounded workers only for concurrency-capable API clients; local clients are serial.
DRES adapter	Implemented transport whose task-specific payloads and complete logging still require live validation. An internal refusal is not an official DRES score.

Metric definitions for a new replay

For N eligible KIS queries, let r_i be the first rank satisfying both the exact video identity and an independently adjudicated target interval. Report the interval/tolerance rule. A miss has reciprocal rank zero:

$$\mathrm{R@k} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{1}[r_i \le k], \qquad \mathrm{MRR} = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{r_i}.$$

Video-only hit rate, temporal hit rate and ANN agreement with exact search are distinct measures and must not share an ambiguous recall column. Missing media, errors and unanswerable cases require explicit eligibility and outcome rules, not post-hoc exclusion to improve rates.

VQA. Score the production answer, not a separately regenerated answer. Exact match uses Unicode NFKC, case folding and whitespace normalization against accepted references; it does not use substring overlap, captions or target-video rank. Report answered/abstained/error counts and coverage over all queries. For answerable set P, answered subset A and exact-match subset C, report $|C|/|P|$, $|A|/|P|$ and $|C|/|A|$; errors and abstentions remain in P, and empty denominators are undefined. Report unanswerable-question outcomes separately. Unsupported-answer rates require independent media-based support labels and an explicit labeled-answer denominator; without labels they are undefined, not zero. Retain source/frame identity, crop coordinates, raw response and runtime configuration for each outcome.

Interaction and AVS. Treat each KIS-C dialogue as one session, preserving questions, oracle/operator replies and ranks per turn. AVS needs relevant-shot labels or judge outcomes, with unique videos/ranges and duplicate rate reported separately; ordinary KIS recall is not a substitute.

Runtime. Separate query processing, retrieval, fusion, VLM calls and end-to-end latency. Record warm-up, trial counts, failures, timeouts and percentile calculation. Throughput counts completed operations of one stated type under a declared workload, not a mixture of candidate and query rates.

Reporting. Freeze the query and corpus manifests, model/provider and index revisions, hardware and raw per-query outputs. Use uncertainty appropriate to each denominator and paired observations for ablations. Do not combine heterogeneous retrieval, VQA, safety and speed measures into an undocumented overall score.